

THE STARSHIP GALILEO

DEEP SPACE TRAVEL AND COLONIZATION

INSTRUCTIONS MANUAL

THE GEMINI PROTOCOLS

PHASE 262

THE QUINN RADIATION UNIFORM - PERSONAL PROTECTION

the suit stack's shield, minus everything else
overlap at every closure — the drizzle finds no gap



event shield on the body, chronic shield in the walls

BISMUTH RUBBER · THE BUNKER PAIRING · THE CHEST TAG

A GARMENT WORN IS WORTH TWO ON THE HOOK

Personal protection — v1.0 in force

GP-IM-262

Revision v1.0 — 24 August 2026

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SEATED LAW — EXTERNAL VET PENDING

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LICENSE: CERN OPEN HARDWARE LICENCE V2 (CERN-OHL-W-2.0)

THE GEMINI PROTOCOLS — CHAPTER 262

PHASE 262: THE QUINN RADIATION UNIFORM - PERSONAL PROTECTION — THE CROSSING UNIFORM — STATUS: CURRENT — SEATED LAW / EXTERNAL VET PENDING (THE 261-265 AUDIT BLOCK WAS ANNOUNCED AT THE CLOSE OF VET BATCH 37 AND NEVER SEATED). The suit stack's shield, minus vacuum, ballistics and thermal — the subtraction is the design. The crossing uniform is the pure radiation garment: bismuth-rubber shield elements in a wearable stack for the event, not the voyage. Event shield versus chronic shield, clean division: solar-particle-event protons are the crossing target, and the uniform is the event answer while the chronic answer stays architectural (260's bricks, 259's tanks). The bunker pairing closes the event doctrine: essentials work in uniforms; non-essentials ride the event in the bunker. Overlap is the closure law: jacket over pants at the waist, hood over collar, flaps over shoe tops — the drizzle finds no gap. The laundry flag is affirmed with the file's own machinery: the wringer line would destroy the shield elastomer — hand-clean with approved agents. Wearable mass is the real constraint: bismuth rubber is dense; the thickness trade is seated — heavier eats more dose, lighter gets worn longer. A Phase 230 rad tag on the chest of every uniform: the dose ledger rides the garment, read at doffing.

Primary Author & Inventor: Archer Quinn, Aerospace Design Engineer, NPhD — Co-Engineers: Gemini 1 AI, Kimi Chat (the audit). Chapter GP-IM-262, v1.0 — 24 August 2026.

§0 — READ THIS FIRST (HOW TO USE THIS CHAPTER)

STANDARD NOTATION — MATERIALS & CALIBRATIONS (series law, seated 19 August 2026, sitting 461): Materials or calibrations is not missing science or engineering — it is, to any real physicist or engineer, stating the fkn obvious. Everyone will use different bench test methods, equipment, materials and weight-to-strength choices, where chemical, magnetic or conduction issues are resolved by choice — e.g. carbon fibre over steel. Our designs are base calibrations to existing strength materials: a rim speed does not exceed current basic materials strength. We do not spec methods, and we do not teach welders how to weld.

Section map: §0 this page — §1 the uniform law as seated (contents line, the uniform in crayon, the physics affirmed, the honest flags, the bench items) — §2 the physics, graded — §3 the system in the uniform — §4 the doctrine record — §6 build sequence — §7 cross-phase — §8 do-not-build — §9 owed — §10 status and provenance.

§1 — THE LAW (AS SEATED, VERBATIM)

THE CONTENTS LINE (verbatim): 'Phase 262: **Radiation Uniform - Personal Protection — the crossing**

uniform. The 92/93 suit stack stripped of vacuum, ballistics, and thermal: the bismuth-silicate shield as clothing (jacket, pants, soft hood — no helmet; shoe-top flap covers, no special shoes). Worn during known radiation crossings and as out-of-work wear on Moon bases and zero-atmosphere planets; the uniform for all Earth-based test labs. Event doctrine: essentials work in the uniform, non-essentials ride the event in the water-tank bunker room. Event shield vs chronic shield — no double-paying. Overlap closures at every join (the maze doctrine, worn); hand clean only (the wringer line would kill the elastomer); sweat rotation seated as doctrine. 1526's sixth leg; Chapter 6 of the radiation book.'

THE CORE OBJECTIVE (verbatim): 'Core Objective: a non-ballistic, non-vacuum, no-helmet garment built on the Phase 93 bismuth-silicate shield — worn as the uniform during known radiation crossings and as out-of-work wear on Moon bases and zero-atmosphere planets, and as the standard uniform in all Earth-based test labs.'

Radiation Baseline: NASA shortfall 1526 (Radiation Countermeasures — Crew and Habitat). Sixth leg of the flesh wound; third cornerstone of the radiation book — the personal layer.

Live instruction, 5 August 2026 — the personal layer, same day as the bricks and the tanks: the author went back to the suits and stripped them for events.

[*VERBATIM — the garment and the doctrine*]: "Non ballistic, non vacuum , space suits with matching head section no helmet version of the space suits, using the same radiation shielding shall be uniform during known radiation crossings, and out of work gear uniforms on moon bases and zero atmosphere planets. all non essentials shall be in the water tank bunker room for those minutes or hours."

[*VERBATIM — the lab law and the cuts*]: "this shall be the uniform for all earth based test labs, pants and jackets are ok, shoes tops can have an attachable flap cover, no need for special shoes"

[*VERBATIM — the spec order and the hygiene law*]: "Go back to the suits and spec everything not related to vacuum, ballistics or temp control as the crossing uniforms. not to be worn in general as they wont work in the laundry hand clean only, and you would stink from the body sweat."

[*VERBATIM — the name and filing*]: "radiation uniform - personal protection"

THE UNIFORM IN CRAYON (VERBATIM)

Take the moon suit and throw away everything that fights the sky: no vacuum bladder, no armor, no heating sandwich. What is left is the shield itself, cut as clothes — a jacket, pants, and a soft hood instead of a helmet, all carrying the bismuth rubber that eats the rays. It is the hospital X-ray apron rebuilt as a uniform you can work in.

It is an event garment. When the ship or the base knows a radiation crossing is coming — a solar storm window, a belt transit — the rule splits the crew: the people who must keep working pull on the uniform, and everyone else goes and sits in the water-tank bunker room for those minutes or hours. The bunker is

the Phase 259 tanks racked as a room; the uniform is the bunker you can walk in.

On a Moon base it hangs by the door as out-of-work wear. In Earth test labs it is simply the lab uniform — the same garment, the same shield, no special shoes: an attachable flap covers the shoe tops, because radiation reads gaps, not fashion.

And it is not pajamas. The shield rubber cannot go through the laundry — hand clean only — and a rubber garment makes you sweat. So it is worn for crossings and off-shift cover, not all day every day. It hangs on its hook, clean and ready, like a firefighter's turnout gear.

THE PHYSICS (AUDIT: AFFIRMED) (VERBATIM)

- **The subtraction is the design.** Removing vacuum, ballistics, and thermal from the 92/93 stack leaves the pure radiation core: the bismuth-silicate elastomer as a garment. No pressure joints, no helmet seals, no thermal sandwich — mass and cost drop to the shield layer alone.
- **Event shield vs chronic shield — clean division.** Solar-particle-event protons are the crossing target, and a few g/cm² of garment cuts an SPE dose hard. Chronic GCR it barely dents — and that job already belongs to 259's tanks and 260's bricks. The uniform is the event layer; the architecture is the chronic layer. No double-paying.
- **The bunker pairing closes the event doctrine:** essentials work in uniforms; non-essentials ride the event in the water-tank bunker room. Two layers, one alarm, zero improvisation.
- **Overlap is the closure law (audit addition, seated):** jacket over pants at the waist, hood over collar, flap over shoe tops — every join overlapped, because radiation reads straight seams even on clothing. Same maze doctrine as the brick bond, worn.
- **The laundry flag is affirmed with the file's own machinery:** the wringer line would destroy the shield elastomer — hand clean only is correct, and the hang-at-the-door hygiene doctrine follows from the sweat physics of rubber garments.
- **Wearable mass is the real constraint:** bismuth rubber is dense. A full garment in the 93 spec runs heavy by Earth-lab standards — fine for hours-long crossings, trivial in lunar gravity, and the reason it is an event uniform and not daily wear. Benched below.

THE HONEST FLAGS (VERBATIM)

- **Garment mass budget.** The shield thickness is a trade: heavier eats more dose, lighter gets worn longer. Set the areal density by event class — belt transit vs storm window — and log wear-time limits per mass. Bench item one.
- **The face stays soft.** The hood shields skull and thyroid; eyes and face remain open. For hours-long events with the body shielded this is accepted — chronic exposure belongs to the architecture, not

the garment. Recorded as a conscious acceptance, not an oversight.

- **Sweat is a mission factor, not a joke.** Rubber garments heat the wearer; crossing shifts in uniform want rotation schedules and liners. The author's stink flag is seated as a work-rest doctrine input.
- **Cleaning chemistry.** Hand-clean with agents compatible with silicone and bismuth-silicate rubber — no solvents, no wringers. The wipe-down protocol joins the phase's build sheet.

BENCH ITEMS (VERBATIM)

- Garment areal density per event class (belt transit / storm window / lab duty) and the mass-versus-wear-time curve.
- Closure spec: overlap dimensions at waist, collar, hood, and shoe-top flap; magnetic or hook-and-loop fasteners — glove-friendly, lint-free.
- Sizing and issue: one uniform per crew member, stowed on the Phase 260 hook blocks at the bunker room door.
- Liner layer for sweat management; rotation schedule template for crossing shifts.
- Wipe-down chemistry list: approved cleaners for the shield elastomer.
- A Phase 230 rad tag on the chest of every uniform — the dose ledger rides the garment, read at doffing.

ORIGIN OF THOUGHT — PHASE 262 (VERBATIM)

Same day as the tanks and the bricks, the author completed the stack's bottom layer: go back to the suits, keep only the shield, and make it clothing. Named and filed by his own hand — radiation uniform, personal protection. The crossing doctrine pairs it with the bunker: the people who must work walk inside their shield; everyone else sits inside the water. Chapter 6 of the radiation book.

Why it matters, in plain English: every radiation plan in the industry ends at 'get to the shelter.' This file answers the next question — what about the people who can't stop working? They wear the shelter. And the Earth-lab clause means the garment earns its keep from day one: every test lab on the ground buys the same uniform, which means the crossing uniform is in production before the first ship ever flies.

§2 — WHY IT WORKS (THE PHYSICS, GRADED)

THE SUBTRACTION IS THE DESIGN: removing vacuum, ballistics and thermal from the 92/93 suit stack leaves the pure shield garment — no pressure bladder, no micrometeorite layer, no thermal shell. What remains is exactly the event armor, wearable inside the ship, donnable in seconds, stowed at the bunker door.

EVENT SHIELD VS CHRONIC SHIELD — CLEAN DIVISION: solar-particle-event protons are the crossing target; the uniform eats the spike, the architecture eats the drizzle. The doctrine does not ask a garment to do a building's job, nor a building to do a garment's.

THE BUNKER PAIRING CLOSES THE EVENT DOCTRINE: essentials work in uniforms; non-essentials ride the event in the bunker. The uniform is what makes 'essentials keep working' survivable — the ship does not stop for the storm.

OVERLAP IS THE CLOSURE LAW (audit addition, seated): jacket over pants at the waist, hood over collar, flaps over shoe tops — every seam shingled so the particle finds mass, not gap. Magnetic or hook-and-loop fasteners, glove-blind.

WEARABLE MASS IS THE REAL CONSTRAINT, FACED: bismuth rubber is dense; a full garment in the 93 spec runs heavy by Earth standards, and in 1/6 g wearable. The trade is seated honestly: heavier eats more dose, lighter gets worn longer — and a garment worn is worth two on the hook.

THE LAUNDRY FLAG, AFFIRMED WITH THE FILE'S OWN MACHINERY: the wringer line would destroy the shield elastomer; hand-clean with agents compatible with silicone and bismuth-silicate rubber — no solvents. Sweat is a mission factor, not a joke: crossing shifts in uniform want rotation, and the liner layer manages it.

§3 — THE SYSTEM IN THE UNIFORM

- **The garment:** the 92/93 stack's shield layer, alone — bismuth-rubber elements, no vacuum, no ballistics, no thermal.
- **The division:** event shield on the body, chronic shield in the architecture — each does its own job.
- **The pairing:** essentials work in uniforms; non-essentials ride the storm in the bunker.
- **The closure:** overlap at waist, collar, hood, shoe-tops — the drizzle finds no gap.
- **The tag:** a Phase 230 rad tag on every chest — the dose ledger rides the garment, read at doffing.
- **The trade:** heavier eats more dose, lighter gets worn longer — the thickness spec is the balance.

§4 — THE DOCTRINE RECORD (HOW THE BOOK ALREADY RUNS ON IT)

- The 92/93 suit stack is the parent: the uniform is its shield layer with the other duties subtracted — the subtraction seated as the design itself.
- 260's hook blocks carry the stowage: one uniform per crew member on the bunker room door hooks — the wall doctrine already issued the hardware.
- 230's rad tags close the loop: the dose ledger rides the garment — the calibrated two-channel tags

from the tag phase, read at doffing, logged per crew member.

- 246's medical band is the doctrine cousin: the crossing uniform is crew medicine worn as clothing — prevention by mass, the same philosophy as the envelope's prevention by temperature.

§5 — VALIDATION: the grade and its full record live in the master and the restart.

§6 — BUILD SEQUENCE

- 1. Cut the garment: jacket, pants, hood, shoe flaps — the shield elastomer in the 93 spec, sized per crew member.
- 2. Shingle the closures: waist, collar, hood, shoe-top overlaps — magnetic or hook-and-loop, glove-blind.
- 3. Pin the tag: a Phase 230 rad tag on every chest — the dose ledger rides the garment.
- 4. Write the rotation: crossing-shift schedule and the liner layer for sweat management.
- 5. Stow at the door: one uniform per crew member on the 260 hook blocks at the bunker room door — issue, drill, log.

§7 — CROSS-PHASE (INLINED IN FULL)

- **Phases 92/93 (the suit stack):** the parent — the uniform is the stack's shield with vacuum, ballistics and thermal subtracted.
- **Phase 260 (the shield bricks):** the stowage — uniforms on the hook blocks at the bunker door; the chronic shield the uniform never tries to be.
- **Phase 230 (the rad tags):** the ledger — the chest tag read at doffing; the dose book per crew member.
- **Phase 259 (the shield tanks):** the bunker — the tank-rack shelter the non-essentials ride; the pairing seated.

§8 — DO-NOT-BUILD

- Do NOT ask it to hold air — vacuum is subtracted; the uniform is the event garment, not a suit.
- Do NOT run a gap — overlap at every closure; the drizzle finds the unshingled seam.
- Do NOT machine-wash it — the wringer kills the elastomer; hand-clean with the approved list.
- Do NOT spec it heavier than wearable — the trade is seated: a garment worn is worth two on the

hook.

- Do NOT skip the tag — no tag, no dose ledger; the garment and the book are one issue.

§9 — OWED

OWED: the bench list, whole — garment areal density per event class (belt transit / storm window / lab duty) with the mass-versus-wear-time curve; the closure spec dimensions; sizing and issue; the liner layer; the wipe-down chemistry list; the tag issue. And the external vet, when it lands — the 261-265 block is still owed by the record.

§10 — STATUS / PROVENANCE / CHANGE CONTROL

STATUS: CHAPTER GP-IM-262, v1.0 — CURRENT — SEATED LAW / EXTERNAL VET PENDING (THE 261-265 AUDIT BLOCK WAS ANNOUNCED AT THE CLOSE OF VET BATCH 37 AND NEVER SEATED), seated law, master v2.1 (numerical print order). Built 24 August 2026, third of the 20-chapter 'you get the next 20 ready' shift (the author's order, 24 August 2026, seated in sitting 623 — 'ok ill check and upload, you get the next 20 ready, they didnt slow us down the sped us up'). First version until publication — 'until i publish it, it is still first version, otherwise is just typing' (his law, 22 August 2026).

PROVENANCE — THE STANDING ORDERS, VERBATIM: the town-trip order (seated in sitting 519): 'ok i have to go to town, i have converted them to pdfs now and am uploading, do the next 10 phases and i will read them when i get back' — the order that opened the chapter program; the follow-on orders (seated in sittings 537, 557, 561, 562, 564, 566, 572, 576, 587, 590, 597, 607, 618, 622 and 623): 'ok next 10' / 'ok next ten' / 'all good next 10' / '10 more before go to the old lady for dinner' / 'do another 20 while i away so we clear the 100 mark for the day when i get back' / 'ok next 20' / 'ok next 20, that will take us past gemini' / 'ok another 20 lets keep this train rollin' / 'ok 12 more i will see what i get done in 30 mins to get to the 200' / 'ok i will do last nights now you do the next 20' / 'ok ill do these 20, you do 20 more' / 'ok next 20' / 'ok ill check and upload, you get the next 20 ready, they didnt slow us down the sped us up'.

CHANGE CONTROL: this design changes only on the Author's orders, seated as sittings in the master record. If the master and this chapter ever disagree, THE MASTER WINS. This is a workshop chapter — it carries the current design only; the full change record lives in the master and the restart.

END OF CHAPTER 262